

Problem 1 (18 Marks)

Johnny has $T = 100$ hours per week to allocate between leisure L and work h , where $h = T - L$. His wage is $w = \$20$ per hour and non-labour income is \$200.

(a) **(4 marks)** Write Johnny's weekly budget constraint in terms of leisure L and income C . Identify:

- the intercept on the income axis,
- the intercept on the leisure axis,
- the slope of the budget line.

(b) **(4 marks)** The government introduces a guaranteed income program:

$$G = 600, \quad t = 0.5$$

- Write Johnny's new disposable income equation while benefits are positive.
- Compute the effective net wage.
- Find the break-even level of weekly earnings.
- Compute the corresponding hours worked at the break-even point.

(c) **(3 marks)** Johnny initially works 40 hours per week. After the program is introduced, he works 30 hours.

- Calculate the change in labour supply.
- Explain whether this change is consistent with income and substitution effects.

(d) **(3 marks)** The working-age population is 5,000,000. The labour force participation rate is 70%, and the unemployment rate is 6%.

- Compute the labour force.
- Compute the number of unemployed workers.
- Compute the number of employed workers.

(e) **(4 marks)** Suppose:

- 60,000 unemployed workers find jobs,
- 40,000 employed workers lose jobs and begin searching,
- 30,000 workers leave the labour force.

Compute:

- the new number of unemployed workers,
- the new labour force,
- the new unemployment rate.

Problem 2 (17 Marks)

A firm faces the following labour market conditions:

- Labour demand schedule:

$$W = 50 - 0.5N$$

- Labour supply schedule:

$$W = 10 + 0.5N$$

(a) **(4 marks)** Find the **competitive equilibrium**:

- equilibrium wage,
- equilibrium employment.

(b) **(4 marks)** Suppose the firm is a monopsonist and hires 20 workers.

- Compute the wage paid to workers.
- Compare this wage to the competitive wage.
- Explain why monopsony leads to lower wages.

(c) **(3 marks)** A minimum wage of \$30 is introduced.

- Compute the number of workers willing to work at this wage.
- Compute the number of workers demanded at this wage.
- Determine actual employment.

(d) **(3 marks)** A country admits 400,000 immigrants:

- Economic: 200,000
- Family: 120,000
- Refugees: 80,000
- Compute the percentage share of each category.
- If economic immigrants increase by 10%, compute the new total number of immigrants.

(e) **(3 marks)** Johnny is considering education that costs \$12,000 today and increases earnings by \$3,000 per year for 8 years. The discount rate is 5%.

Use:

$$PV = A \left(\frac{1 - (1 + r)^{-T}}{r} \right)$$

- Compute the present value of benefits.
- Compute the net present value (NPV).
- Should Johnny invest? Briefly explain.